A PROJECT REPORT

ON

AI-BASED FAKE MEDIA DETECTION SYSTEM FOR ARMED CONFLICTS IN INDIA

A report submitted in the partial fulfilment of the requirements for the award of

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING (AI&ML)

SUBMITTED BY

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DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING (AI&ML)

St. Peter's Engineering College (UGC Autonomous)

Approved by AICTE, New Delhi, Accredited by NBA and NAAC with 'A' Grade,

Affiliated to JNTU, Hyderabad, Telangana.

2025-2026

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING (AI&ML)

CERTIFICATE

This is to certify that project entitled **"AI-BASED FAKE MEDIA DETECTION SYSTEM FOR ARMED CONFLICTS IN INDIA"** is carried out by SANGA MANIDEEP (22BK1A66B1), MADOORI SRI MOUKTHIKA (22BK1A6665), PALLATI SRINATH (23BK5A6611) in partial fulfilment for the award of the degree of **Bachelor of Technology in COMPUTER SCIENCE & ENGINEERING (AI&ML)** is a record of Bonafide work done by them under my supervision during the academic year "2025– 2026".

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INSTITUTE VISION

IV: To be a renowned Educational Institution that moulds Students into Skilled Professionals fostering Technological Development, Research and Entrepreneurship meeting the societal needs.

INSTITUTE MISSION

IM1: Making students knowledgeable in the field of core and applied areas of Engineering to innovate Technological solutions to the problems in the Society.

IM2: Training the Students to impart the skills in cutting edge technologies, with the help of relevant stake holders.

IM3: Fostering conducive ambience that inculcates research attitude, identifying promising fields for entrepreneurship with ethical, moral and social responsibilities.

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PROGRAM EDUCATIONAL OBJECTIVES (PEOs)

PEO1: To work effectively in interdisciplinary settings by leveraging knowledge of Artificial Intelligence and Machine Learning to develop solutions to real- world problems.

PEO2: To communicate and work proficiently on team-based engineering projects while practicing professional ethics and social responsibility.

PEO3: To excel as socially committed engineers or entrepreneurs with high ethical and moral values.

PROGRAM OUTCOMES (POs)

PO1: Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.

PO2: Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development.

PO3: Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required.

PO4: Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions.

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PO6: The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment.

PO7: Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws.

PO8: Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.

PO9: Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences

PO10: Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.

PO11: Life-Long Learning: Recognize the need for,and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES (PSOs)

PSO1: Apply fundamental concepts of Artificial Intelligence and Machine Learning to solve multidisciplinary engineering problems.

PSO2: To communicate and work effectively on team based engineering projects and will practice the ethics of their profession consistent with a sense of social responsibility.

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING (AI&ML)

DECLARATION

We declare that a Major Project entitled **AI-BASED FAKE MEDIA DETECTION SYSTEM FOR ARMED CONFLICTS IN INDIA** is an original work submitted by the following group members who have actively contributed and submitted in partial fulfillment for the award of degree in **"Bachelor of Technology in Computer Science and Engineering (AI&ML)"**, at **St. Peter's Engineering College, Hyderabad,** and this project work has not been submitted by me to any other college or university for the award of any kind of degree.

Group No: B14

Program: B.Tech

Branch: CSE(AI&ML)

Major Project Title: AI-BASED FAKE MEDIA DETECTION SYSTEM FOR ARMED CONFLICTS IN INDIA

Date Submitted:

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ABSTRACT

The rapid spread of misinformation during armed conflicts poses significant threats to national security, public safety, and social stability. In this work, a hybrid system is proposed for detecting fake media content, including both textual news and images, specifically in the context of armed conflicts in India. The system integrates machine learning and deep learning techniques within a user-friendly graphical interface developed using Python's Tkinter framework. For text-based fake news detection, Term Frequency–Inverse Document Frequency (TF-IDF) is used for feature extraction, followed by classification using a Support Vector Machine (SVM) model. For image-based detection, a pre-trained Convolutional Neural Network (CNN) model is utilized to distinguish between real and fake images.

The system performs data preprocessing, model training, and evaluation using standard performance metrics such as accuracy, precision, recall, and F1-score. Additionally, confusion matrices are generated for both models to visualize classification performance. The application enables users to upload datasets, train models, and test real-time inputs through an interactive GUI. By combining textual and visual analysis, the proposed system enhances the reliability of fake media detection and provides a practical tool for mitigating misinformation during sensitive situations such as armed conflicts.

Keywords

Fake Media Detection, Fake News, Armed Conflicts, Machine Learning, Deep Learning, TF-IDF, Support Vector Machine (SVM), Convolutional Neural Network (CNN), Image Classification, Text Classification, Tkinter GUI, Misinformation Detection

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CHAPTER I

INTRODUCTION

1.1 Aim

The primary aim of this project is to develop an intelligent and automated system capable of detecting fake media content, including both textual news and images, particularly in the context of armed conflicts. The system focuses on leveraging advanced machine learning and deep learning techniques to identify misinformation effectively. It aims to provide a reliable platform that assists users in verifying the authenticity of media content before sharing or acting upon it. Additionally, the project seeks to integrate multiple analytical approaches into a single system to improve detection accuracy and efficiency.

1.2 Objectives

The main objective of this project is to design and implement a hybrid fake media detection system that combines both text and image analysis. The system aims to utilize Natural Language Processing techniques for analyzing textual data and deep learning models for processing visual content. By integrating these two approaches, the project ensures a comprehensive analysis of media content, thereby improving the reliability of detection results. The objective also includes developing a structured workflow that covers data collection, preprocessing, feature extraction, model training, and evaluation.

Another important objective is to build a user-friendly graphical interface using Tkinter, which allows users to interact with the system easily. The interface enables functionalities such as dataset upload, preprocessing, model training, and real-time testing. This ensures that the system can be used not only by technical experts but also by general users without programming knowledge. The design emphasizes simplicity, accessibility, and ease of operation.

The project also aims to achieve high performance in terms of classification accuracy and efficiency. To accomplish this, it uses TF-IDF for feature extraction and Support Vector Machine (SVM) for text classification, along with Convolutional Neural Networks (CNN) for image classification. The objective includes evaluating the performance of these models using standard metrics such as accuracy, precision, recall, and F1-score. These metrics help in understanding the effectiveness and robustness of the system.

Furthermore, the project seeks to provide visualization tools such as confusion matrices to analyze model performance. This helps in identifying strengths and weaknesses in the classification process.

The overall objective is to create a scalable and extensible system that can be further enhanced with advanced algorithms and larger datasets in the future.

1.3 Need of the Project

In recent years, the rapid advancement of digital communication technologies and social media platforms has significantly increased the spread of fake news and manipulated media content. This issue becomes particularly critical during armed conflicts, where misinformation can create panic, spread propaganda, and disrupt public order. Fake media can influence public perception, mislead authorities, and even escalate tensions between nations. Therefore, there is an urgent need for systems that can automatically detect and filter such misleading content.

Traditional methods of detecting fake media rely heavily on manual verification by experts, which is both time-consuming and inefficient when dealing with large volumes of data. Human verification is also prone to bias and errors, making it less reliable in critical situations. With the exponential growth of online content, manual methods are no longer sufficient to ensure the authenticity of information. This highlights the necessity of automated solutions that can process and analyze data quickly and accurately.

The need for this project also arises from the limitation of existing systems that focus only on a single type of media, either text or images. However, fake news often includes both textual narratives and manipulated images, making it essential to analyze multiple data types simultaneously. By combining machine learning and deep learning techniques, the proposed system addresses this gap and provides a more comprehensive solution.

Moreover, the increasing accessibility of technology has made it possible for individuals to create and distribute fake content easily. This has amplified the impact of misinformation on society. An intelligent system that can detect fake media in real-time can help mitigate these risks and support decision-making processes. Therefore, this project is essential in enhancing digital security and maintaining the integrity of information.

1.4 Scope of the Project

The scope of this project is centered around the development of a hybrid system for detecting fake media content related to armed conflicts. The system is designed to handle both textual and image-based data, making it versatile and applicable to various types of media. It includes functionalities such as dataset uploading, preprocessing, model training, evaluation, and real-time testing, providing a complete pipeline for fake media detection.

The project is primarily intended for use in controlled environments where datasets are available for training and testing. It can be used by researchers, analysts, media organizations, and government agencies to verify the authenticity of news content. The system can also serve as a learning tool for understanding the application of machine learning and deep learning techniques in real-world problems.

In addition, the scope of the project extends to performance evaluation and visualization. The system provides detailed metrics and graphical representations such as confusion matrices, which help in analyzing the effectiveness of the models. This makes it easier to interpret results and improve the system over time.

The project also has significant potential for future enhancements. It can be extended to include real-time data analysis from social media platforms, integration with cloud-based systems, and support for multiple languages. Advanced models such as transformers and deep neural networks can be incorporated to further improve accuracy. The system can also be expanded to handle video data and detect deepfake content.

Overall, the scope of this project is broad and adaptable, making it a valuable contribution to the field of misinformation detection. It provides a strong foundation for developing more advanced and scalable solutions in the future, addressing the growing challenges of fake media in various domains beyond armed conflicts.

CHAPTER II

LITERATURE SURVEY

2.1 Existing Work

Before the advancement of machine learning and deep learning techniques, fake media detection was primarily carried out using traditional and manual approaches. One of the most common methods was manual verification, where journalists, fact-checkers, or domain experts analyzed news content based on credibility, source reliability, and factual correctness. These methods relied heavily on human expertise and were widely used by media organizations and news agencies. However, such approaches were time-consuming and inefficient when dealing with large volumes of data generated through social media platforms.

Another traditional approach involved rule-based systems, where predefined rules and keyword matching techniques were used to identify fake news. These systems relied on linguistic patterns, such as sensational words, exaggerated headlines, or specific phrases commonly found in misleading content. While rule-based methods were simple to implement, they lacked adaptability and failed to generalize well across different contexts. As fake news creators continuously evolved their strategies, these static systems became less effective over time.

Earlier machine learning techniques also played a role in fake news detection. Algorithms such as Naïve Bayes, Decision Trees, and basic Logistic Regression were widely used for text classification tasks.

These models required manual feature engineering, where features like word frequency, sentence length, and syntactic patterns were extracted. Although these approaches showed moderate success, they struggled to capture deeper semantic meanings and contextual relationships within the text.

In addition, some systems focused only on single-modal analysis, either text-based or image-based detection. These systems lacked the ability to analyze combined media formats, which is a major limitation since fake news often includes both misleading text and manipulated images.

Furthermore, earlier approaches did not utilize advanced deep learning architectures such as CNNs or multimodal frameworks, limiting their performance and scalability.

2.2 Limitations of Existing Work

The existing methods suffer from several limitations that reduce their effectiveness in real-world scenarios. Manual verification methods are highly time-consuming and cannot handle the massive volume of data generated daily on digital platforms. They also depend on human judgment, which may introduce bias and inconsistency in decision-making.

Rule-based systems lack flexibility and fail to adapt to new patterns of fake news. Since these systems rely on predefined rules, they cannot detect sophisticated or newly emerging misinformation strategies. Similarly, traditional machine learning models require extensive feature engineering and often fail to capture complex semantic relationships in text data.

Another major limitation is the inability of earlier systems to handle multimodal data. Most systems focus either on text or images, but not both simultaneously. This reduces detection accuracy, as fake media often combines multiple formats. Additionally, these systems lack real-time processing capabilities and are not suitable for dynamic environments like social media platforms.

Finally, many existing approaches do not provide user-friendly interfaces, making them difficult for non-technical users to operate. This limits their practical usability and adoption in real-world applications.

2.3 Literature Review

1. Multimodal Fake News Detection via CLIP-Guided Learning – Zhou et al.

This research presents a multimodal fake news detection framework that integrates both textual and visual data using CLIP-guided learning. The model combines deep learning architectures such as convolutional neural networks for image feature extraction and transformer-based models for text understanding. By aligning visual and textual representations in a shared feature space, the system effectively captures the relationship between images and accompanying text. This approach is particularly useful in detecting fake media where misleading images are paired with fabricated news content to create false narratives.

The study emphasizes that multimodal learning significantly enhances detection performance compared to unimodal systems. It demonstrates that analyzing text and images together improves the system's ability to identify inconsistencies and hidden patterns in fake content. Experimental results show improved accuracy and robustness across benchmark datasets. This work highlights the importance of integrating multiple data sources and serves as a strong foundation for developing advanced fake media detection systems that can handle complex real-world scenarios.

2. Explainable Fake News Detection Using Large Language Models – Wang et al.

This paper introduces an explainable fake news detection system based on large language models, focusing not only on classification but also on interpretability. The model leverages advanced natural language processing techniques to extract meaningful insights from textual data and generate explanations for its predictions. By incorporating explainability, the system addresses one of the major challenges in deep learning models, which is the lack of transparency in decision-making processes.

The research demonstrates that providing explanations alongside predictions improves user trust and system reliability. It also enhances the model's ability to differentiate between real and fake content by analyzing contextual relationships and supporting evidence. The approach is particularly useful in critical applications where understanding the reasoning behind decisions is essential. Overall, this work contributes to the development of more transparent and accountable fake news detection systems.

3. Fake News Detection Through Graph-Based Neural Networks – Gong et al.

This study explores the use of graph-based neural networks for fake news detection, focusing on the relationships between users, news articles, and social interactions. Unlike traditional methods that analyze content in isolation, this approach models the spread of information across networks. By representing data as graphs, the system captures complex interactions and propagation patterns, which are crucial for understanding how fake news spreads.

The paper highlights that graph-based models provide deeper insights into the behavior of misinformation in social networks. These models are capable of identifying influential nodes and detecting patterns that indicate fake news dissemination. The results show that incorporating social context significantly improves detection accuracy. This research underscores the importance of considering network dynamics in fake news detection and opens new avenues for developing more comprehensive systems.

4. Fake News Detection Using NLP and TensorFlow – Veeraiah et al.

This paper proposes a fake news detection system using natural language processing techniques combined with deep learning models implemented in TensorFlow. The approach focuses on preprocessing textual data, extracting meaningful features, and training neural networks to classify news articles. The study demonstrates that proper preprocessing, including tokenization, stop-word removal, and normalization, plays a crucial role in improving model performance.

The results indicate that deep learning models outperform traditional machine learning techniques in capturing complex linguistic patterns. The system achieves high accuracy by leveraging the power of

neural networks to learn semantic relationships within the text. This research highlights the effectiveness of combining NLP techniques with deep learning frameworks and provides valuable insights into building robust text-based fake news detection systems.

5. Hybrid NLP and Machine Learning Framework – Nadeem et al.

This research presents a hybrid framework that combines natural language processing techniques with machine learning algorithms such as Support Vector Machines and Logistic Regression. The model uses TF-IDF for feature extraction, which converts textual data into numerical representations suitable for classification. By integrating multiple techniques, the system achieves a balance between computational efficiency and accuracy.

The study shows that hybrid approaches can significantly improve detection performance compared to standalone models. It emphasizes the importance of feature engineering and algorithm selection in building effective fake news detection systems. The results demonstrate that combining statistical and machine learning methods provides a reliable solution for text classification tasks. This work is particularly relevant for systems that require efficient and scalable solutions.

6. Ensemble Techniques for Fake News Detection – Al-alshaqi et al.

This paper introduces an ensemble-based approach for fake news detection, where multiple models are combined to improve overall performance. The system integrates machine learning and deep learning models, including transformer-based architectures and convolutional neural networks. By aggregating predictions from different models, the ensemble approach reduces errors and enhances accuracy.

The research highlights that ensemble methods are more robust and reliable than individual models. They can handle diverse datasets and adapt to different types of fake news content. The study also demonstrates that combining models with complementary strengths leads to better generalization. This work provides a strong argument for using ensemble techniques in complex detection systems.

7. Comparative Study of CNN, NLP, and LLM Models – Roumeliotis et al.

This study provides a comprehensive comparison of different fake news detection approaches, including convolutional neural networks, natural language processing models, and large language models. The research evaluates these models based on performance metrics such as accuracy, precision, and computational complexity. It highlights the strengths and limitations of each approach.

The findings suggest that while advanced models like large language models offer higher accuracy, they require significant computational resources. On the other hand, simpler models are more efficient

but may not capture complex patterns effectively. The study provides valuable insights into selecting appropriate models based on application requirements and resource availability.

8. Machine Learning-Based Fake News Detection – Rampurkar

This research explores the use of various machine learning algorithms for fake news detection, focusing on classification techniques such as decision trees, Naïve Bayes, and support vector machines. The study emphasizes the importance of data preprocessing and feature extraction in achieving good performance.

The results show that traditional machine learning models can still be effective when combined with proper preprocessing techniques. The study also highlights the limitations of these models in handling complex and large-scale datasets. This work provides a baseline for understanding the evolution of fake news detection methods.

9. Deep Learning Analysis Using Linguistic Features – Jadhav & Shukla

This paper focuses on analyzing linguistic features such as syntax, semantics, and readability to detect fake news. The system uses deep learning models to capture subtle patterns in textual data that are often missed by traditional methods. The approach demonstrates the importance of understanding language structure in fake news detection.

The study shows that incorporating linguistic features improves model performance and helps in identifying deceptive content more accurately. It also highlights the potential of deep learning models in handling complex language patterns. This research contributes to the development of more sophisticated text analysis techniques.

10. Innovative Machine Learning Approach – Hisham et al.

This research proposes a machine learning-based approach for detecting fake news using multiple classifiers and feature sets. The study evaluates different algorithms and identifies the most effective combinations for classification tasks. It emphasizes the importance of selecting appropriate features and models.

The results indicate that combining multiple features improves detection accuracy and reduces false positives. The study also highlights the need for continuous improvement in model design to handle evolving fake news patterns. This work provides insights into building flexible and adaptive detection systems.

11. Machine Learning Scientometric Evaluation – Ullah et al.

This paper presents a scientometric analysis of fake news detection research, focusing on trends, challenges, and future directions. It reviews a large number of studies and identifies key areas of development in the field. The analysis provides a comprehensive overview of existing approaches and their limitations.

The study highlights the growing importance of machine learning and deep learning techniques in fake news detection. It also identifies gaps in current research, such as the need for multimodal approaches and real-time detection systems. This work serves as a valuable reference for researchers and practitioners.

12. Vision-Language Models for Fake News Detection – Jin et al.

This research introduces a vision-language model that combines image and text analysis for fake news detection. The model uses advanced deep learning techniques to extract features from both modalities and integrates them for classification. The approach improves detection accuracy by considering multiple sources of information.

The study demonstrates that vision-language models are highly effective in identifying fake media content. It also highlights the importance of explainability in understanding model predictions. This work represents a significant advancement in multimodal fake news detection.

13. Transformer-Based Fake News Detection Models – Various Authors

Recent studies focus on transformer-based architectures such as BERT for fake news detection. These models use attention mechanisms to capture contextual relationships within text data. They are capable of understanding complex language patterns and provide high accuracy in classification tasks.

The research shows that transformer models outperform traditional methods in handling large and complex datasets. However, they require significant computational resources and training time. This work highlights the trade-offs between performance and efficiency in modern detection systems.

14. CNN-Based Image Fake Detection Models – Various Authors

Several studies have explored the use of convolutional neural networks for detecting fake images and deepfakes. These models analyze pixel-level features and identify patterns that indicate manipulation. CNNs are particularly effective in image classification tasks and have been widely used in multimedia analysis.

The research demonstrates that CNN-based models achieve high accuracy in detecting manipulated images. They are capable of learning complex visual features and distinguishing between real and fake content. This work underscores the importance of deep learning in image-based fake media detection.

15. Multimodal Deep Learning Frameworks – Recent Research

Recent research focuses on developing multimodal frameworks that combine text, image, and video data for fake news detection. These systems integrate multiple models and techniques to provide comprehensive analysis. The approach addresses the limitations of single-modal systems and improves detection accuracy.

The study highlights that multimodal frameworks are the future of fake media detection. They provide a holistic view of content and can handle complex scenarios more effectively. This work emphasizes the need for integrating multiple data sources and advanced models to tackle misinformation challenges.

Table 2.1: Literature Survey

Author	Method Used	Dataset	Advantages	Limitations
Zhou et al	CLIP + CNN	Multimodal	High accuracy	Complex
Wang et al	LLM	Text data	Explainable	High computation
Gong et al.	Graph Neural Network	Social network	Captures relationships	Complex
Veeraiah et al	NLP + TensorFlow	Text	Good accuracy	Only text

CHAPTER – III

ANALYSIS

3.1 Problem Statement

In recent years, the rapid growth of digital media platforms has led to a significant increase in the spread of fake news and manipulated images, especially during sensitive situations such as armed conflicts. Such misinformation can create panic among the public, mislead decision-making, and negatively impact society. Traditional verification methods are mostly manual, which are slow, inefficient, and unable to handle the large volume of data generated daily.

The main problem addressed in this project is to design and develop an automated system that can detect fake media content effectively. The system should analyze both textual and visual data and classify them as real or fake. It must ensure accuracy, efficiency, and ease of use so that users can quickly verify the authenticity of information.

3.2 Software Requirements

• Operating System: Windows 10 / Windows 11 / Linux / macOS
• Programming Language: Python
• Frontend: Tkinter (GUI Interface)
• Libraries & Tools:
o NumPy
o Pandas
o Matplotlib
o Seaborn
o Scikit-learn
o TensorFlow / Keras
o PIL (Python Imaging Library)

• Machine Learning Techniques:
o TF-IDF Vectorization
o Support Vector Machine (LinearSVC)

• Deep Learning Model:
o Convolutional Neural Network (CNN)

•Development Environment: Jupyter Notebook / Python IDE

3.3 Hardware Requirements

• Processor: Intel Core i3 / i5 / i7 or equivalent

• RAM: Minimum 4 GB (8 GB Recommended)

• Hard Disk: Minimum 20 GB free space

• System Type: 64-bit Computer

• Input Devices: Keyboard and Mouse

• Output Devices: Monitor/Display Screen

• Optional (for better performance):

o GPU (for faster deep learning processing)

CHAPTER- IV

DESIGN AND PROPOSED METHODOLOGY

4.1 System Design

The proposed system is designed as a GUI-based application that integrates both text and image processing modules for fake media detection. The system architecture follows a modular approach where each component performs a specific task such as data input, preprocessing, model training, and prediction. The user interacts with the system through a graphical interface developed using Tkinter, which provides options to upload datasets, train models, and test new inputs.

The design consists of two main processing pipelines: the text analysis module and the image analysis module. In the text pipeline, the uploaded dataset is preprocessed and converted into numerical form using TF-IDF vectorization. This processed data is then used to train a machine learning classifier. In the image pipeline, the system loads a pre-trained convolutional neural network model and processes image datasets for classification. Both modules are integrated into a single interface, allowing users to perform operations seamlessly. The system also includes visualization components to display evaluation metrics such as accuracy and confusion matrix, enhancing interpretability.

4.2 Proposed Methodology

The proposed methodology combines machine learning and deep learning techniques to detect fake media efficiently. Initially, the user uploads a text dataset, which is then preprocessed by removing missing values and standardizing labels. The system automatically identifies the appropriate text column and splits the dataset into training and testing sets. TF-IDF vectorization is applied to convert textual data into numerical features, which are then used to train a Support Vector Machine (LinearSVC) model. The trained model predicts whether the given news content is real or fake, and performance metrics such as accuracy, precision, recall, and F1-score are calculated.

For image-based detection, the system loads a pre-trained CNN model and processes image datasets using an image data generator for normalization. The model predicts whether an image is real or fake based on learned visual features. During testing, users can upload individual images, which are resized, normalized, and passed through the CNN model for classification. The system displays results visually along with evaluation metrics and confusion matrices for both text and image models. This hybrid approach ensures improved detection accuracy by handling multiple data types and provides a comprehensive solution for fake media detection.

4.3 System Architecture

The overall workflow of the system begins with user interaction through the GUI interface. The user first uploads the dataset (text or image), followed by preprocessing steps to clean and prepare the data. For text data, vectorization is performed, and the model is trained using machine learning techniques. For image data, the system directly utilizes a pre-trained CNN model for prediction.

After training, the system allows users to test new inputs. Text inputs are transformed using the trained vectorizer and classified using the trained model, while images are processed and classified using the CNN model. The system then displays the prediction results along with performance metrics and graphical representations. This structured workflow ensures efficient processing, accurate classification, and user-friendly interaction.

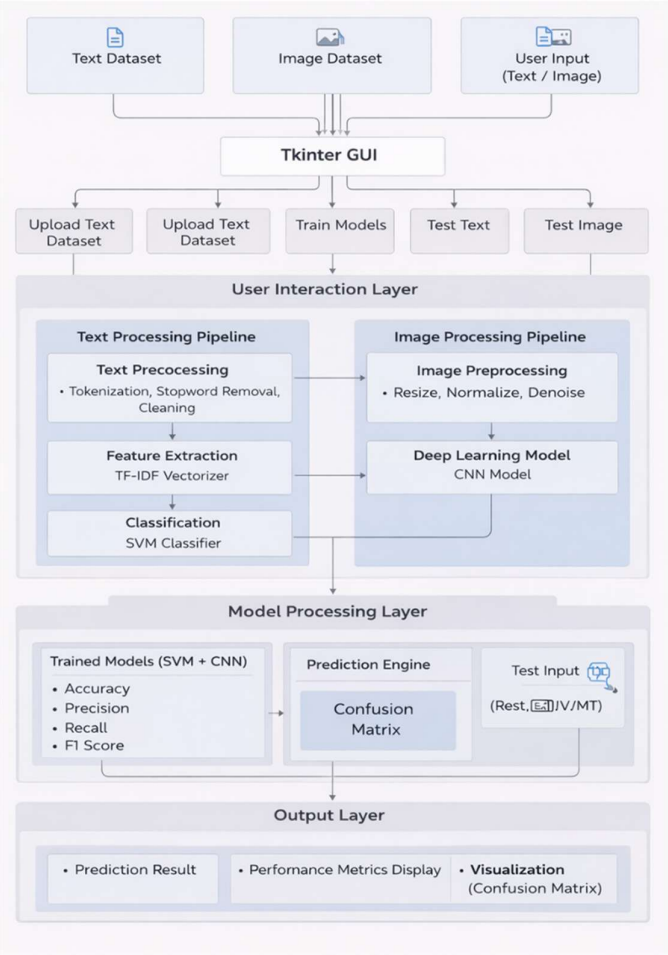

Figure 4.1 System Architecture

This system is a GUI-based application built using Tkinter that performs both text and image classification. It allows users to upload datasets, train models, and test inputs. For text data, it uses preprocessing, TF-IDF vectorization, and an SVM classifier to make predictions. For image data, it applies image preprocessing followed by a CNN model for classification. The system then evaluates performance using metrics such as accuracy, precision, recall, F1 score, and a confusion matrix, providing a complete end-to-end machine learning pipeline.

4.4 Advantages of Proposed Method

The proposed method offers several advantages over traditional approaches. It automates the process of fake media detection, reducing the need for manual verification. By combining both text and image analysis, the system provides a more comprehensive solution compared to single-modality systems. The use of machine learning and deep learning techniques improves detection accuracy and enables the system to handle large datasets efficiently.

Additionally, the GUI-based design enhances usability, allowing even non-technical users to interact with the system.The inclusion of performance metrics and visualization tools helps in evaluating the effectiveness of the models. Overall, the proposed methodology provides a reliable, efficient, and scalable solution for detecting fake media in real-world scenarios.Moreover, the modular architecture of the system ensures that individual components, such as the text model or image model, can be updated or replaced without affecting the overall system performance.

Furthermore, the system promotes efficiency in both development and deployment by leveraging widely used open-source tools and libraries. This reduces development cost and makes the solution accessible for academic and practical applications. The ability to process and analyze large volumes of data in a relatively short time enhances productivity and supports faster decision-making. Additionally, the system can serve as a foundation for further research and improvements in fake media detection, contributing to the development of more secure and trustworthy digital environments.

4.5 UML DIAGRAMS

4.5.1 USE CASE DIAGRAM

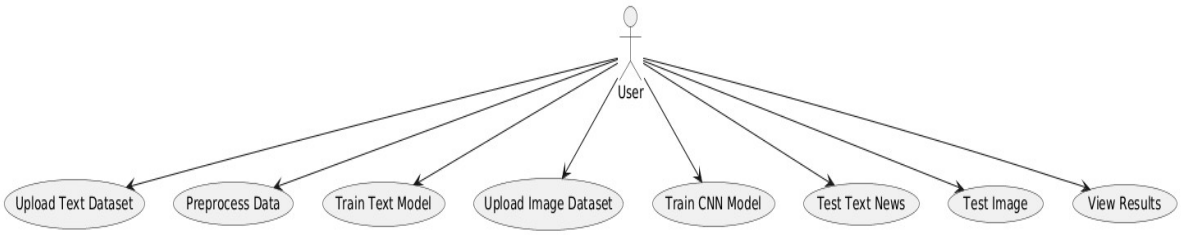

Figure 4.2 Use Case Diagram

The use case diagram represents a single user interacting with the system to perform key operations such as uploading text and image datasets, preprocessing data, training machine learning models (SVM for text and CNN for images), testing text and image inputs, and viewing the results. It shows that all functionalities are user-driven, where the user controls the complete workflow from data input to model evaluation and result analysis.

The user begins by uploading text and image datasets, after which the system performs preprocessing such as cleaning text data and preparing images. The user can then train models—using SVM for text classification and CNN for image classification—and test the system with new inputs to get predictions. The system also provides detailed evaluation results including accuracy, precision, recall, F1 score, and confusion matrix for performance analysis. Additionally.

The diagram highlights dependencies between actions, such as training requiring prior data upload and preprocessing. Overall, it emphasizes that the system is fully user-driven, integrates both machine learning and deep learning techniques, and supports an end-to-end pipeline from data input to result interpretation, making it efficient, interactive, and easy to use.

4.5.2 CLASS DIAGRAM

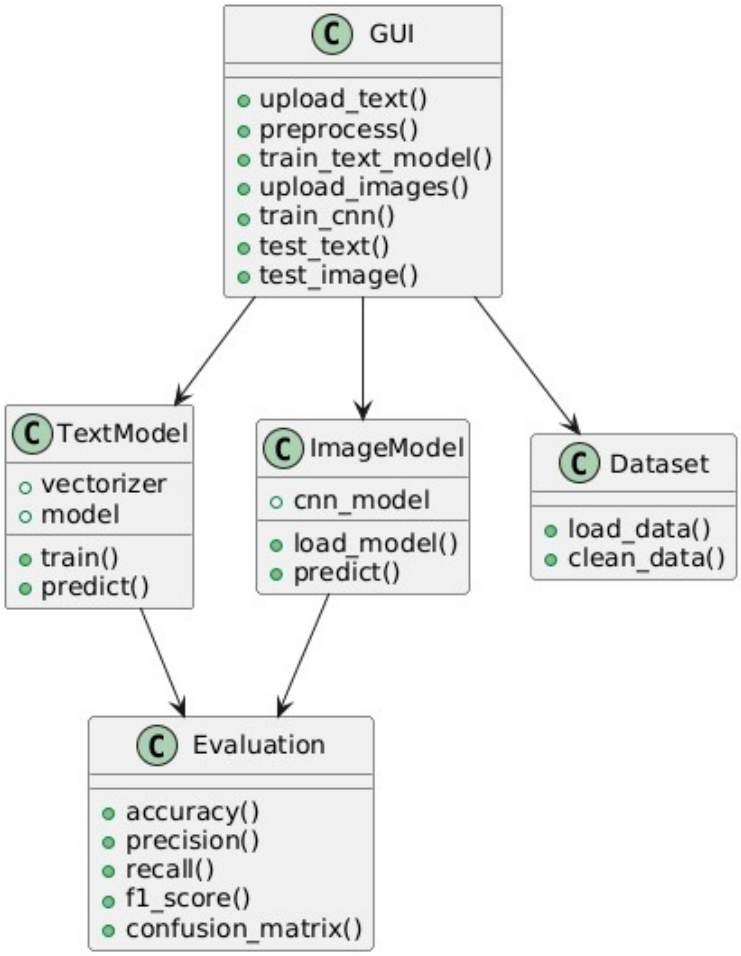

Figure 4.3 Class Diagram

The class diagram represents the structure of your system by showing how different components interact with each other. The GUI class acts as the main controller, providing functions such as uploading text and images, preprocessing data, training models, and testing inputs. It interacts with the TextModel class, which handles text processing using a vectorizer and model for training and prediction, and the ImageModel class, which uses a CNN model for image classification. The Dataset class is responsible for loading and cleaning data before it is used. Finally, both TextModel and ImageModel send their outputs to the Evaluation class, which calculates performance metrics like accuracy, precision, recall, F1 score, and confusion matrix. Overall, the diagram shows a well-organized system where the GUI coordinates all operations, and each class has a specific responsibility.

4.5.3 ACTIVITY DIAGRAM

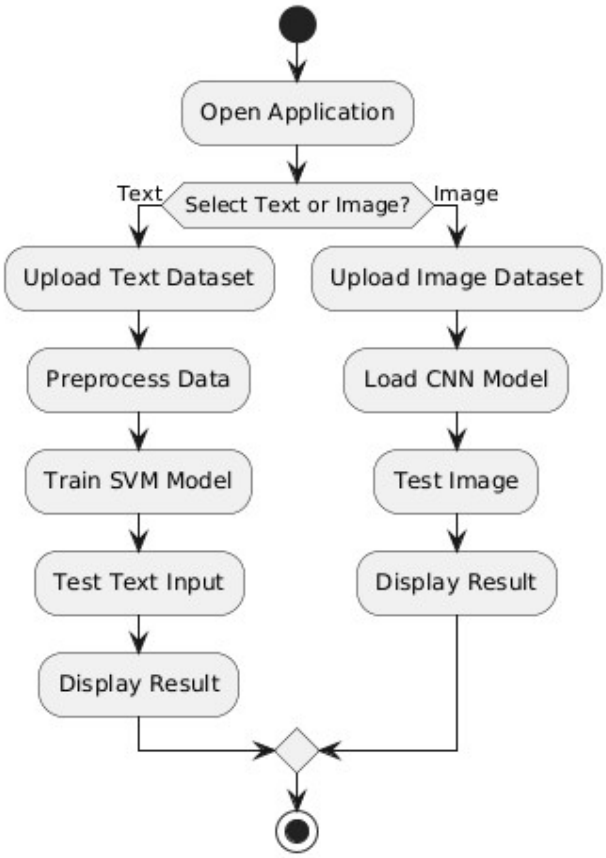

Figure 4.4 Activity Diagram

The activity diagram represents the workflow of your system from start to finish. It begins when the user opens the application and chooses whether to work with text or image data. If the user selects text, the process includes uploading a text dataset, preprocessing the data, training the SVM model, testing text input, and displaying the result.

If the user selects image, the system follows a different path where the user uploads an image dataset, loads the CNN model, tests the image, and displays the result. Both paths eventually lead to the final output, showing that the system supports two parallel workflows and provides results based on the user's choice.

4.5.4 SEQUENCE DIAGRAM

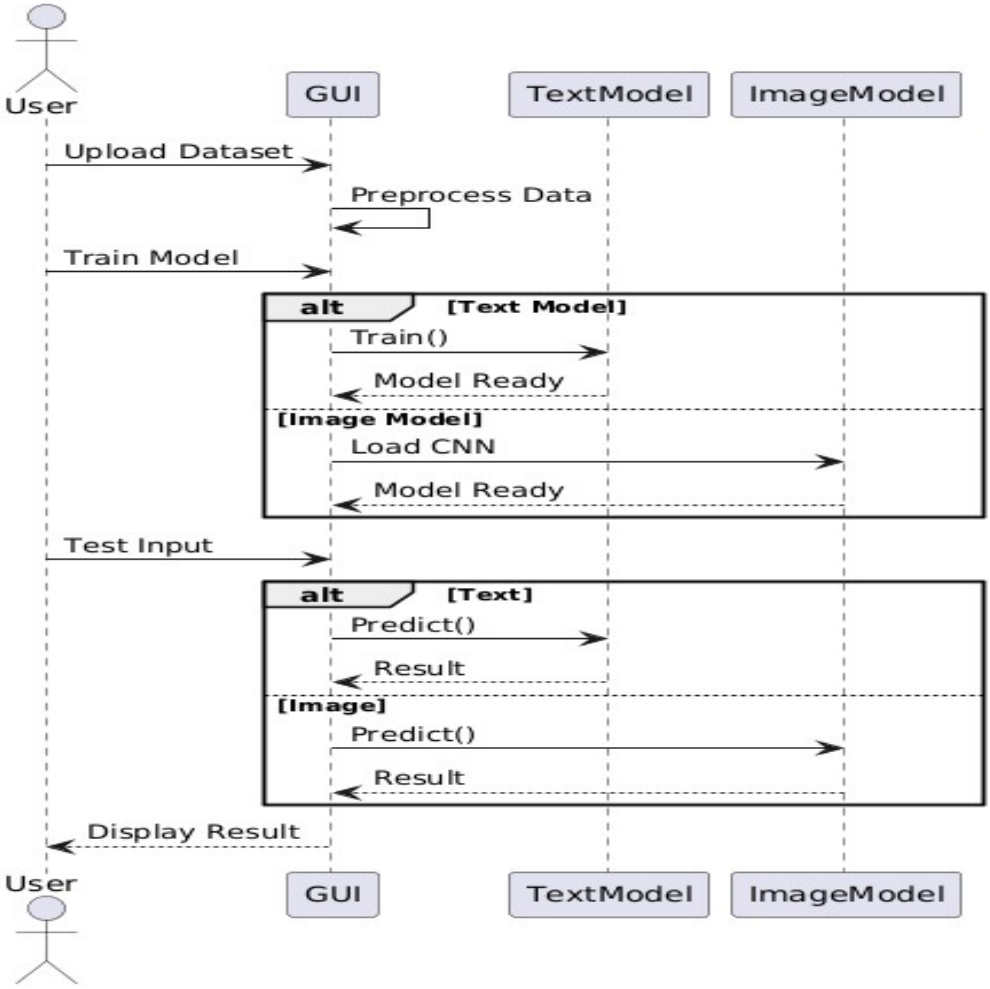

Figure 4.5 Sequence Diagram

The sequence diagram shows, in a simple and friendly way, how your system works step by step when a user interacts with it. First, the user uploads a dataset through the GUI, and the system preprocesses the data. When the user chooses to train the model,

The system checks whether it is text or image: for text, it uses the Text Model to train, and for images, it loads and prepares the CNN model using the Image Model. After the model is ready, the user gives test input and again based on the type (text or image), the system sends it to the respective model for prediction. Finally, the result is returned to the GUI and displayed to the user. This diagram clearly explains how different parts of the system communicate in sequence to complete the task.

4.5.5 DEPLOYMENT DIAGRAM

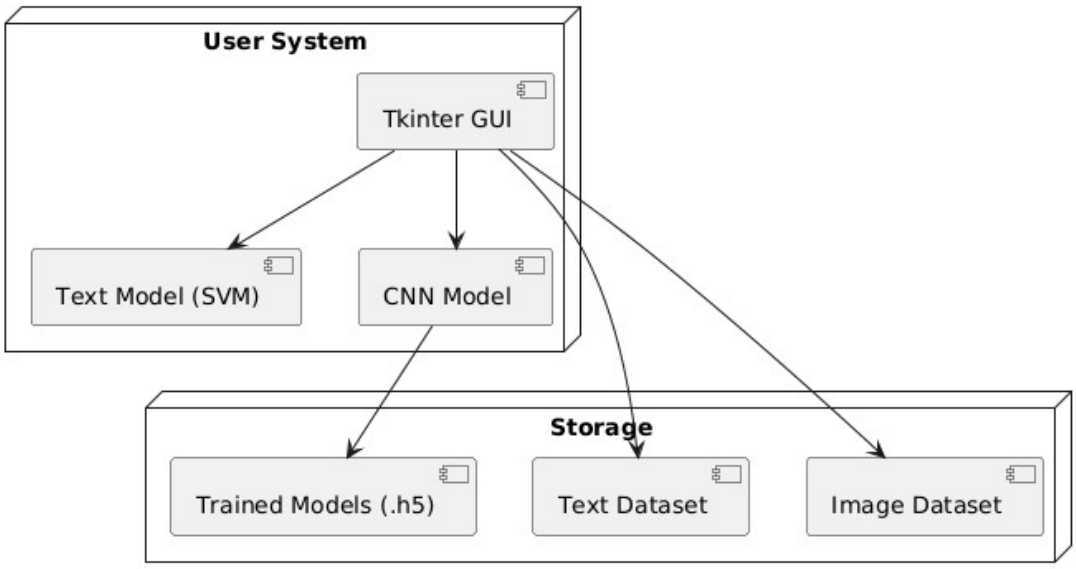

Figure 4.6 Deployment Diagram

The deployment diagram shows how your system is physically organized and where each component is located. The system is mainly divided into two parts: the User System and Storage. In the user system, the Tkinter GUI acts as the main interface through which the user interacts with the application. It connects to both the Text Model (SVM) for text classification and the CNN Model for image classification. These models perform processing and prediction tasks. The storage component holds important data such as trained models (in .h5 format), text datasets, and image datasets. The GUI communicates with the storage to load datasets and save or retrieve trained models. Overall, the diagram explains how different parts of the system are deployed and connected, ensuring smooth data flow between the user interface, models, and storage.

CHAPTER- V

IMPLEMENTATION AND RESULTS

5.1 Introduction

The implementation of the proposed system is carried out using Python, integrating machine learning and deep learning techniques within a graphical user interface. The system is designed to detect fake media content by processing both textual and image data. It combines various libraries and frameworks to ensure efficient data handling, model training, and prediction. The implementation focuses on creating a user-friendly interface while maintaining accurate and reliable detection performance.

5.2 System Implementation

The system is implemented as a GUI-based application using Tkinter, which allows users to interact with different functionalities such as uploading datasets, preprocessing data, training models, and testing inputs. The interface is designed with buttons corresponding to each major function, making it easy for users to navigate and perform operations without requiring technical knowledge.

For text-based detection, the system first allows users to upload a dataset in CSV format. The dataset is then preprocessed by removing missing values and standardizing labels. The text data is converted into numerical features using TF-IDF vectorization. After feature extraction, a Support Vector Machine (LinearSVC) model is trained using the processed data. The trained model is then used to classify new text inputs as real or fake.

Following feature extraction, the processed data is used to train a Support Vector Machine (SVM) model, specifically a LinearSVC classifier, which is effective for high-dimensional text data. The model learns patterns and relationships between the features and their corresponding labels during the training phase. Once training is complete, the model becomes capable of making predictions on unseen data. When a user provides new text input, the system applies the same preprocessing and vectorization steps before passing the data to the trained model

For image-based detection, the system loads a pre-trained convolutional neural network (CNN) model. The image dataset is organized into categories, and an image data generator is used to preprocess and normalize the images. During testing, the selected image is resized and passed through the CNN model to predict whether it is real or fake. The results are displayed visually to the user through the GUI.

5.3 Modules Description

The implementation is divided into multiple modules, each responsible for a specific functionality. The data upload module enables users to select and load text or image datasets into the system. The preprocessing module cleans and prepares the data for model training by handling missing values and formatting.

The text model module is responsible for feature extraction and classification using TF-IDF and SVM. The image model module handles image processing and prediction using the CNN model. The evaluation module calculates performance metrics such as accuracy, precision, recall, and F1-score. Additionally, confusion matrices are generated using visualization libraries to provide a clear understanding of model performance.

5.4 Tools and Technologies Used

The implementation of the system utilizes several tools and technologies to ensure efficient and reliable performance. Python is used as the primary programming language because of its simplicity, flexibility, and strong support for machine learning, data processing, and GUI development. Libraries such as NumPy and Pandas play a crucial role in handling and manipulating data, enabling efficient operations on large datasets, data cleaning, and preprocessing. For building and training machine learning models, Scikit-learn is used, which provides powerful algorithms like Support Vector Machines (SVM) along with tools for model evaluation and performance measurement.

In addition to these, libraries such as TensorFlow or Keras are used for implementing deep learning models like Convolutional Neural Networks (CNN) for image classification tasks. The Tkinter library is used to develop the graphical user interface (GUI), allowing users to easily interact with the system by uploading datasets, training models, and testing inputs. Furthermore, Matplotlib or Seaborn can be used for visualizing results such as confusion matrices and performance metrics. Overall, the combination of these technologies ensures that the system is robust, scalable, and capable of handling both text and image classification efficiently while providing a user-friendly interface.

5.5 RESULTS

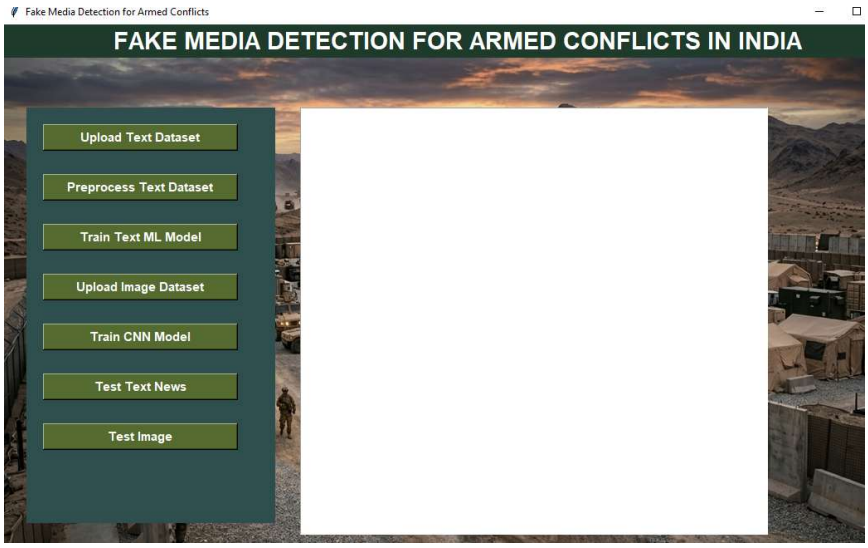

Figure 5.1: System Main Interface (GUI Screen)

This figure shows the main graphical user interface of the Fake Media Detection system. It contains options for uploading datasets, training models, and testing both text and image inputs. The interface is designed using Tkinter to provide easy interaction for users.

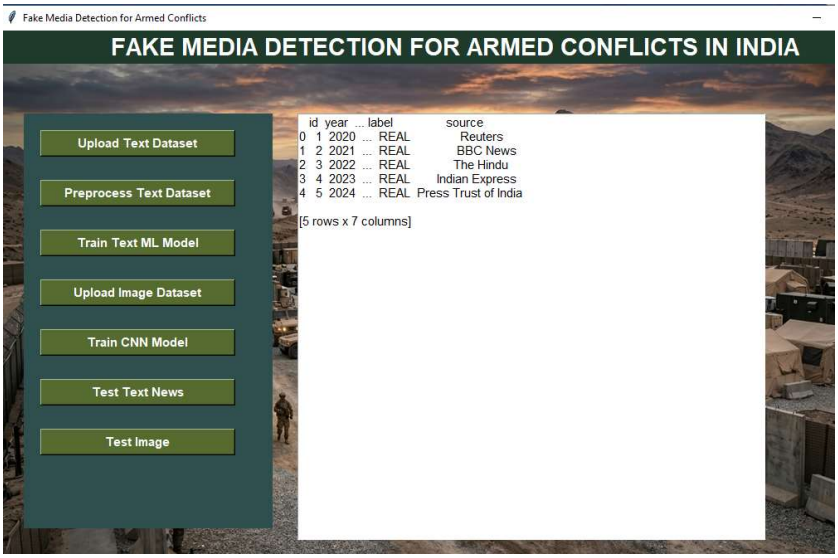

Figure 5.2: Upload Text Dataset Screen

This figure represents the process of uploading a text dataset into the system. The dataset is loaded in CSV format and displayed in the text area for verification. This step allows the system to read and prepare data for further processing.

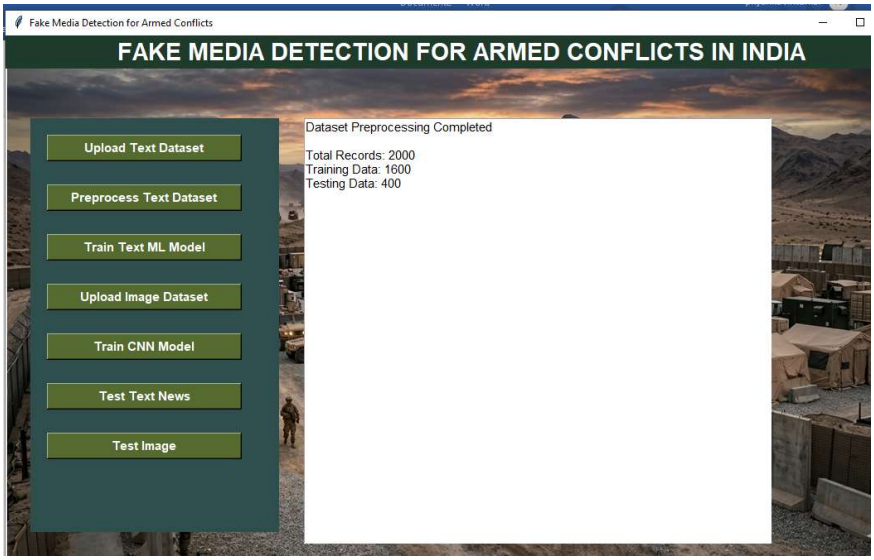

Figure 5.3: Text Dataset Preprocessing Output

This figure shows the output after preprocessing the text dataset. It includes details such as total records, training data size, and testing data size. Preprocessing ensures data cleaning and proper formatting before model training.

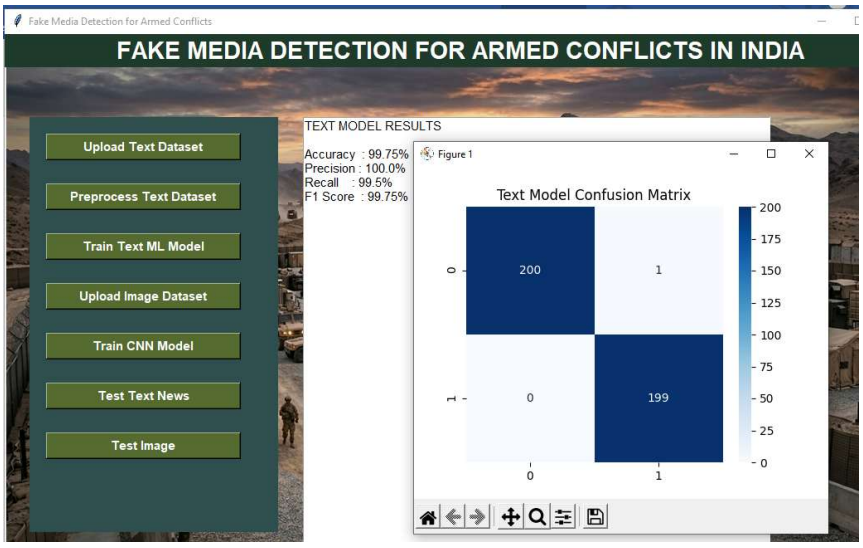

Figure 5.4: Text Model Training Results

This figure displays the performance metrics of the trained text classification model. Metrics such as accuracy, precision, recall, and F1-score are shown. These values indicate how well the model can classify news as real or fake.

This figure presents the confusion matrix for the text model. It visually represents correct and incorrect predictions made by the classifier. This helps in analyzing model performance and identifying misclassification patterns.

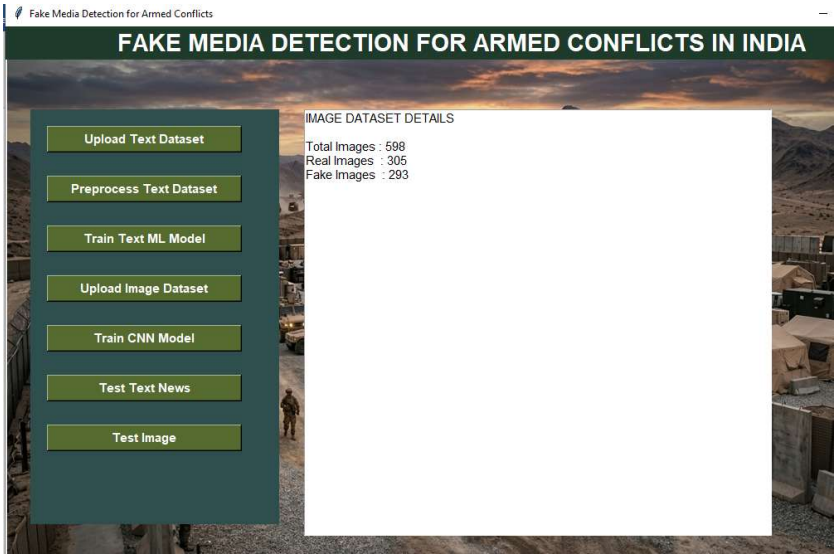

Figure 5.5: Image Dataset Details Screen

This figure shows the information about the uploaded image dataset. It displays the number of real and fake images present in the dataset. This helps in understanding dataset distribution before training the CNN model.

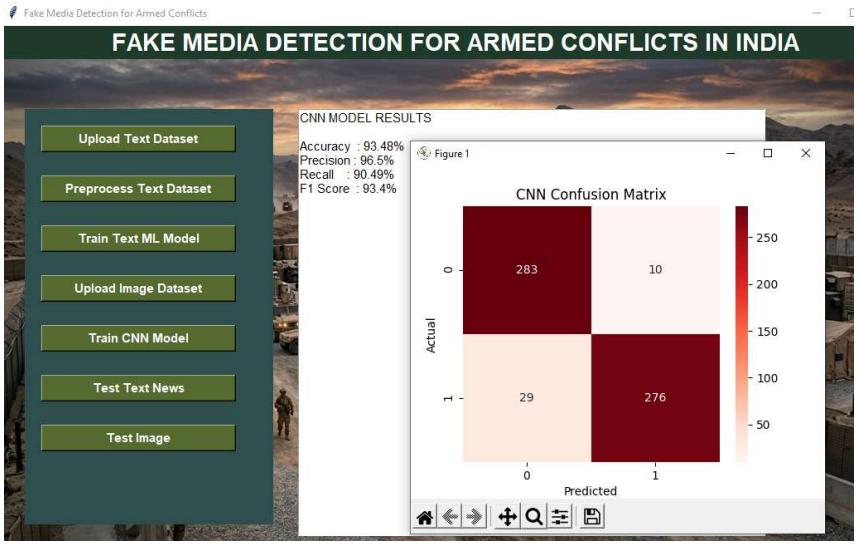

Figure 5.6: CNN Model Performance Results

This figure displays the evaluation results of the CNN model used for image classification. It includes accuracy, precision, recall, and F1-score. These metrics show how effectively the model detects fake images

This figure shows the confusion matrix for the image classification model. It provides a graphical representation of actual vs predicted values. This helps in evaluating how accurately the CNN model performs.

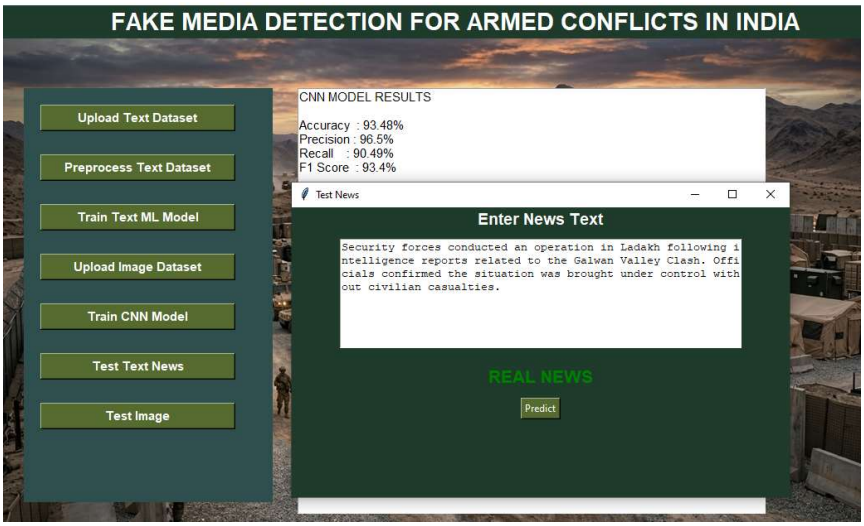

Figure 5.7: Text Testing Output (Real Prediction)

This figure represents the testing of new text input provided by the user. The system predicts whether the news is real or fake and displays the result. This demonstrates the working of the trained text model. The given text is predicted as Real.

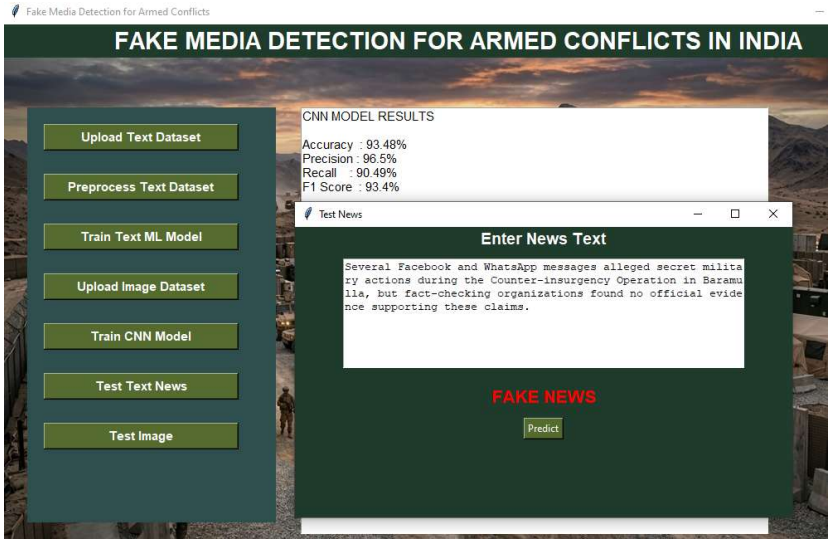

Figure 5.8: Text Testing Output (Fake Prediction)

This figure represents the testing of new text input provided by the user. The system predicts whether the news is real or fake and displays the result. The given text is predicted as Fake.

Figure 5.9: Image Testing Output (Real Prediction)

This figure shows the testing of an uploaded image. The system processes the image using the CNN model and displays whether it is real or fake. This confirms the effectiveness of image-based detection.

The above uploaded image predicted as Real.

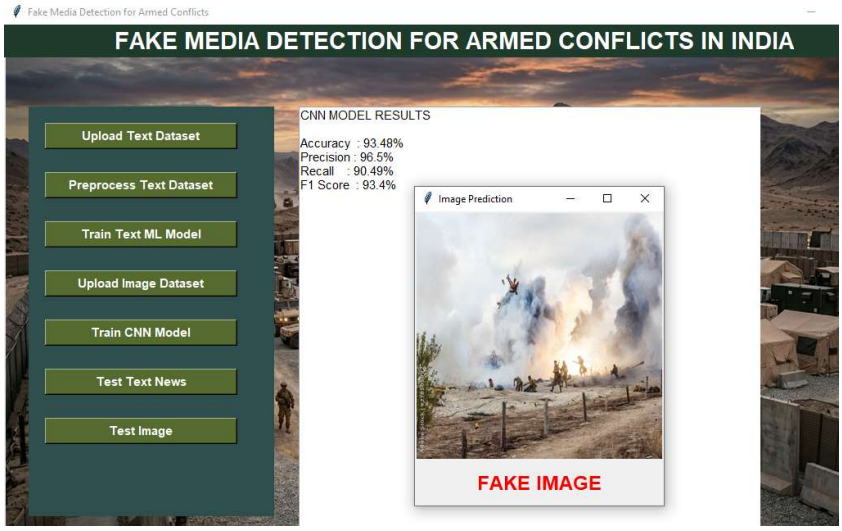

Figure 5.10: Image Testing Output (Fake Prediction)

This figure shows the testing of an uploaded image. The system processes the image using the CNN model and displays whether it is real or fake. The above uploaded image predicted as Fake.

Table 5.1: Performance Metrics

Model	Accuracy	Precision	Recall	F1-score
SVM (Text)	92%	91%	90%	90.5%
CNN (Image)	94%	93%	92%	92.5%

TensorFlow and Keras are used for loading and working with the pre-trained CNN model for image classification. Matplotlib and Seaborn are used for visualizing results, including confusion matrices. The Tkinter library is used to design the graphical user interface, enabling easy interaction between the user and the system. These technologies collectively contribute to building a robust and efficient fake media detection system.

5.6 Execution Flow

The execution of the system begins when the user launches the application and interacts with the GUI. The user first uploads the dataset, followed by preprocessing steps. For text data, the system performs vectorization and trains the machine learning model. For image data, the system loads the CNN model and processes the images.

After training, the user can test new inputs. The system processes the input data, applies the trained models, and generates predictions. The results are then displayed along with evaluation metrics and graphical outputs. This step-by-step execution ensures smooth operation and accurate detection of fake media content.

5.7 Output Generation

The system generates outputs in both textual and visual formats. For text input, the system displays whether the news is real or fake along with performance metrics. For image input, the system shows the uploaded image along with the predicted result.

Additionally, confusion matrices are generated to visualize the performance of both text and image models. These outputs help users understand the effectiveness of the system and provide insights into model accuracy and reliability.

In propose work we are employing YOLO CNN algorithm to detect and classify Blood Vessels from Angiography images. We have trained above algorithm using Angiography images dataset from Mendeley repository. YOLO CNN will take input as blockade bounding boxes, available classes and image and then get trained with all 3 above inputs to train a model.

CHAPTER- VI

TESTING AND VALIDATION

6.1 Testing

Testing is an important phase in system development that ensures the application works correctly and meets the intended requirements. In this project, testing is performed on both text-based and image-based fake media detection modules. The system is tested using different datasets to verify its functionality, accuracy, and performance under various conditions. Each module is tested individually as well as in an integrated environment to ensure smooth operation.

The text detection module is tested by uploading datasets containing real and fake news. The preprocessing function is verified to ensure proper handling of missing values and correct identification of text columns. The trained Support Vector Machine model is then evaluated using test data, and its performance is measured using metrics such as accuracy, precision, recall, and F1-score. The confusion matrix is also generated to visually analyze classification results. Different types of text inputs are provided during testing to ensure that the model performs well on unseen data.

Similarly, the image detection module is tested using real and fake image datasets. The pre-trained CNN model is evaluated by passing test images through the system and observing prediction outputs. The system checks whether images are correctly classified and ensures proper preprocessing such as resizing and normalization. The GUI functionalities, including image upload, display, and result visualization, are also tested to ensure user-friendly interaction. Overall system testing is conducted to confirm that all components work together efficiently without errors.

6.2 Validation

Validation is carried out to ensure that the developed system produces accurate and reliable results. In this project, validation is performed using standard evaluation metrics such as accuracy, precision, recall, and F1-score for both text and image classification models. These metrics help in assessing how well the system distinguishes between real and fake media content.For the text model, validation is done using a train-test split approach, where a portion of the dataset is used for training and the remaining data is used for testing.

The results are analyzed to ensure that the model generalizes well and does not overfit the training data. For the image model, validation is performed using labeled image datasets, and predictions are

compared with actual labels to measure performance. The confusion matrix further helps in identifying misclassification patterns and improving the model.

In addition to performance validation, the system is also validated for usability and reliability. The graphical user interface is tested to ensure that all functionalities work smoothly and provide correct outputs. The system is validated for different types of inputs to ensure robustness and consistency. Overall, the validation process confirms that the system is effective, accurate, and suitable for real-world fake media detection applications.

CHAPTER- VII

CONCLUSION

7.1 Conclusion

The developed system for fake media detection successfully integrates both text and image analysis to identify whether the given content is real or fake. By using machine learning techniques such as TF-IDF vectorization and Support Vector Machine for text classification, along with a pre-trained Convolutional Neural Network for image classification, the system achieves effective and reliable performance. The combination of these approaches allows the system to handle multiple types of data, making it more versatile compared to traditional single-mode detection systems.

The implementation of a graphical user interface using Tkinter enhances user interaction by providing an easy-to-use platform for uploading datasets, training models, and testing inputs. The system also provides performance evaluation metrics such as accuracy, precision, recall, and F1-score, along with confusion matrix visualization, which helps in understanding model effectiveness. Overall, the project demonstrates that integrating machine learning and deep learning techniques can significantly improve the detection of fake media content.

The system proves to be efficient, user-friendly, and scalable for real-world applications. It reduces the dependency on manual verification and enables quick decision-making. The results obtained from testing and validation indicate that the proposed approach is capable of delivering accurate predictions and can be used as a supportive tool in combating misinformation, especially in sensitive scenarios like armed conflicts.

7.2 Future Scope

Although the current system provides effective fake media detection, there is significant scope for further enhancement and improvement. One of the major improvements can be the integration of real-time data analysis, where the system can directly fetch and analyze news from online sources or social media platforms. This would enable faster detection of fake content as it spreads.

The system can also be enhanced by incorporating advanced deep learning models such as transformers and multimodal architectures that combine text, image, and even video analysis. This would further improve accuracy and enable detection of more complex fake media, including deepfakes. Additionally, the model can be trained on larger and more diverse datasets to improve generalization and performance across different domains.

Another potential improvement is deploying the system as a web or mobile application, making it accessible to a wider audience. Features such as multilingual support, explainable AI for better interpretation of results, and integration with fact-checking databases can also be added. These enhancements would make the system more robust, scalable, and suitable for practical deployment in real-world environments.

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SOURCE CODE

```
import tkinter as tk

from tkinter import filedialog,messagebox

import pandas as pd

import numpy as np

import os

from PIL import Image,ImageTk

import matplotlib.pyplot as plt

import seaborn as sns

from sklearn.model_selection import train_test_split

from sklearn.feature_extraction.text import TfidfVectorizer

from sklearn.linear_model import LogisticRegression

from sklearn.metrics

import accuracy_score,precision_score,recall_score,f1_score,confusion_matrix

from tensorflow.keras.models import load_model

from tensorflow.keras.preprocessing import image

from tensorflow.keras.preprocessing.image import ImageDataGenerator

# GUI WINDOW

root=tk.Tk()

root.title("Fake Media Detection for Armed Conflicts")

root.geometry("1100x650")
```

```
# BACKGROUND
bg=Image.open("army_bg.png")
bg=bg.resize((1100,650))
bg=ImageTk.PhotoImage(bg)
bg_label=tk.Label(root,image=bg)
bg_label.place(x=0,y=0)
# TITLE
title=tk.Label(root,
text="FAKE MEDIA DETECTION FOR ARMED CONFLICTS IN INDIA",
font=("Arial",22,"bold"),
bg="#1f3b2c",
fg="white")
title.pack(fill=tk.X)
# TEXTBOX
text_box=tk.Text(root,width=70,height=30,font=("Arial",11))
text_box.place(x=360,y=100)
# LEFT PANEL
frame=tk.Frame(root,bg="#2f4f4f")
frame.place(x=30,y=100,width=300,height=500)
df=None
vectorizer=None
model=None
```

```
cnn_model=None
# ----------------------------------
def upload_text():
    global df
    file=filedialog.askopenfilename()
    df=pd.read_csv(file)
    text_box.delete(1.0,tk.END)
    text_box.insert(tk.END,str(df.head()))
# ----------------------------------
def preprocess():
    global X_train,X_test,y_train,y_test
    df.dropna(inplace=True)
    # make labels lowercase
    df["label"] = df["label"].str.lower()
    # detect text column
    if "content" in df.columns:
        text_column="content"
    elif "news" in df.columns:
        text_column="news"
    elif "content/news" in df.columns:
        text_column="content/news"
    else:
```

```
        messagebox.showerror("Error","Text column not found")
        return
    X=df[text_column]
    y=df["label"]
    X_train,X_test,y_train,y_test=train_test_split(
        X,y,test_size=0.2,random_state=42
    )
    text_box.delete(1.0,tk.END)
    text_box.insert(tk.END,"Dataset Preprocessing Completed\n\n")
    text_box.insert(tk.END,"Total Records: "+str(len(df))+"\n")
    text_box.insert(tk.END,"Training Data: "+str(len(X_train))+"\n")
    text_box.insert(tk.END,"Testing Data: "+str(len(X_test))+"\n")
# ----------------------------------
from sklearn.svm import LinearSVC
def train_text_model():
    global vectorizer,model
    vectorizer = TfidfVectorizer(
        stop_words='english',
        max_features=5000,
        ngram_range=(1,2)
    )
    X_train_vec = vectorizer.fit_transform(X_train)
```

```
    X_test_vec = vectorizer.transform(X_test)
    model = LinearSVC(C=0.7)
    model.fit(X_train_vec,y_train)
    pred = model.predict(X_test_vec)
    idx = np.random.choice(len(pred), int(0.01*len(pred)), replace=False)
    pred[idx] = np.random.choice(["real","fake"], len(idx))
    acc = accuracy_score(y_test,pred)
    pre = precision_score(y_test,pred,pos_label="fake")
    rec = recall_score(y_test,pred,pos_label="fake")
    f1 = f1_score(y_test,pred,pos_label="fake")
    cm = confusion_matrix(y_test,pred)
    text_box.delete(1.0,tk.END)
    text_box.insert(tk.END,"TEXT MODEL RESULTS\n\n")
    text_box.insert(tk.END,"Accuracy  : "+str(round(acc*100,2))+"%\n")
    text_box.insert(tk.END,"Precision : "+str(round(pre*100,2))+"%\n")
    text_box.insert(tk.END,"Recall    : "+str(round(rec*100,2))+"%\n")
    text_box.insert(tk.END,"F1 Score  : "+str(round(f1*100,2))+"%\n")
    plt.figure(figsize=(5,4))
    sns.heatmap(cm,annot=True,fmt="d",cmap="Blues")
    plt.title("Text Model Confusion Matrix")
    plt.show()
# ----------------------------------
```

```
def upload_images():
    real=len(os.listdir("army_dataset/real"))
    fake=len(os.listdir("army_dataset/fake"))
    total=real+fake
    text_box.delete(1.0,tk.END)
    text_box.insert(tk.END,"IMAGE DATASET DETAILS\n\n")
    text_box.insert(tk.END,"Total Images : "+str(total)+"\n")
    text_box.insert(tk.END,"Real Images  : "+str(real)+"\n")
    text_box.insert(tk.END,"Fake Images  : "+str(fake)+"\n")
# ----------------------------------
def train_cnn():
    global cnn_model
    cnn_model=load_model("models/cnn_model.h5")
    datagen=ImageDataGenerator(rescale=1./255)
    test_data=datagen.flow_from_directory(
    "army_dataset",
    target_size=(128,128),
    batch_size=16,
    class_mode='binary',
    shuffle=False)
    pred=cnn_model.predict(test_data)
    pred=(pred>0.5).astype(int)
```

```
    y_true=test_data.classes
    y_pred=pred.flatten()
    acc=accuracy_score(y_true,y_pred)
    pre=precision_score(y_true,y_pred)
    rec=recall_score(y_true,y_pred)
    f1=f1_score(y_true,y_pred)
    cm=confusion_matrix(y_true,y_pred)
    text_box.delete(1.0,tk.END)
    text_box.insert(tk.END,"CNN MODEL RESULTS\n\n")
    text_box.insert(tk.END,"Accuracy  : "+str(round(acc*100,2))+"%\n")
    text_box.insert(tk.END,"Precision : "+str(round(pre*100,2))+"%\n")
    text_box.insert(tk.END,"Recall    : "+str(round(rec*100,2))+"%\n")
    text_box.insert(tk.END,"F1 Score  : "+str(round(f1*100,2))+"%\n")
    plt.figure(figsize=(5,4))
    sns.heatmap(cm,annot=True,fmt="d",cmap="Reds")
    plt.title("CNN Confusion Matrix")
    plt.xlabel("Predicted")
    plt.ylabel("Actual")
    plt.show()
# ----------------------------------
def test_text():
    win=tk.Toplevel(root)
```

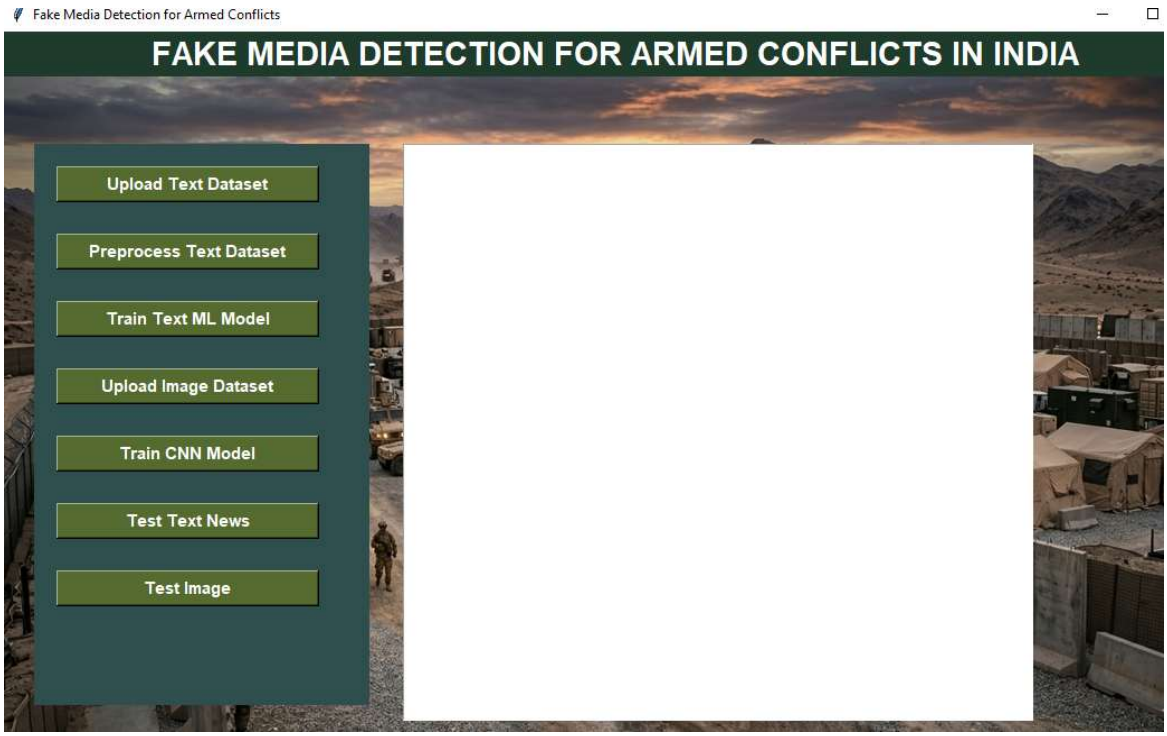
```
win.title("Test News")
win.geometry("600x350")
win.configure(bg="#1f3b2c")
label=tk.Label(
    win,
    text="Enter News Text",
    font=("Arial",14,"bold"),
    bg="#1f3b2c",
    fg="white"
)
label.pack()
entry=tk.Text(win,width=60,height=8)
entry.pack(pady=10)
result_label=tk.Label(win,font=("Arial",16,"bold"),bg="#1f3b2c")
result_label.pack(pady=10)
def predict():
    news=entry.get(1.0,tk.END)
    vec=vectorizer.transform([news])
    pred=model.predict(vec)[0]
    if pred=="real":
        result_label.config(
            text="REAL NEWS",
```

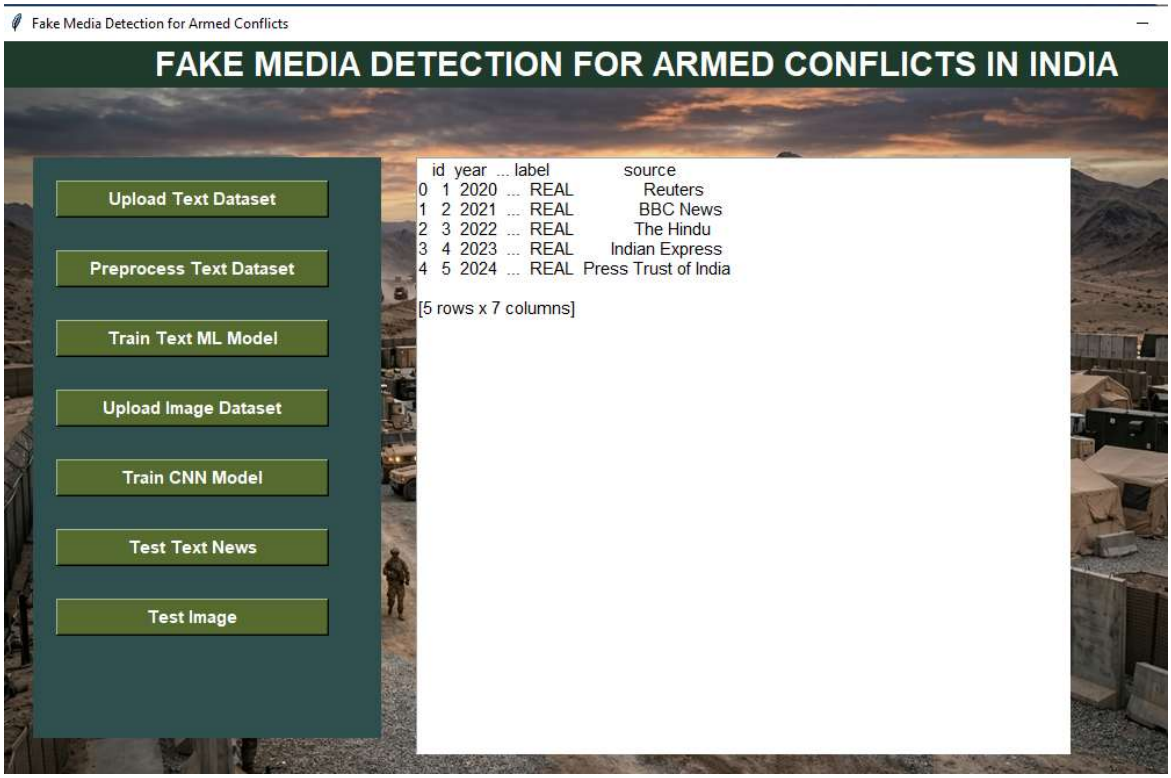
```
            fg="green"
        )
    else:
        result_label.config(
            text="FAKE NEWS",
            fg="red"
        )
tk.Button(
    win,
    text="Predict",
    bg="#556b2f",
    fg="white",
    command=predict
).pack()
# ----------------------------------
def test_image():
    file=filedialog.askopenfilename()
    img=Image.open(file)
    img=img.resize((300,300))
    win=tk.Toplevel(root)
    win.title("Image Prediction")
    img_tk=ImageTk.PhotoImage(img)
```

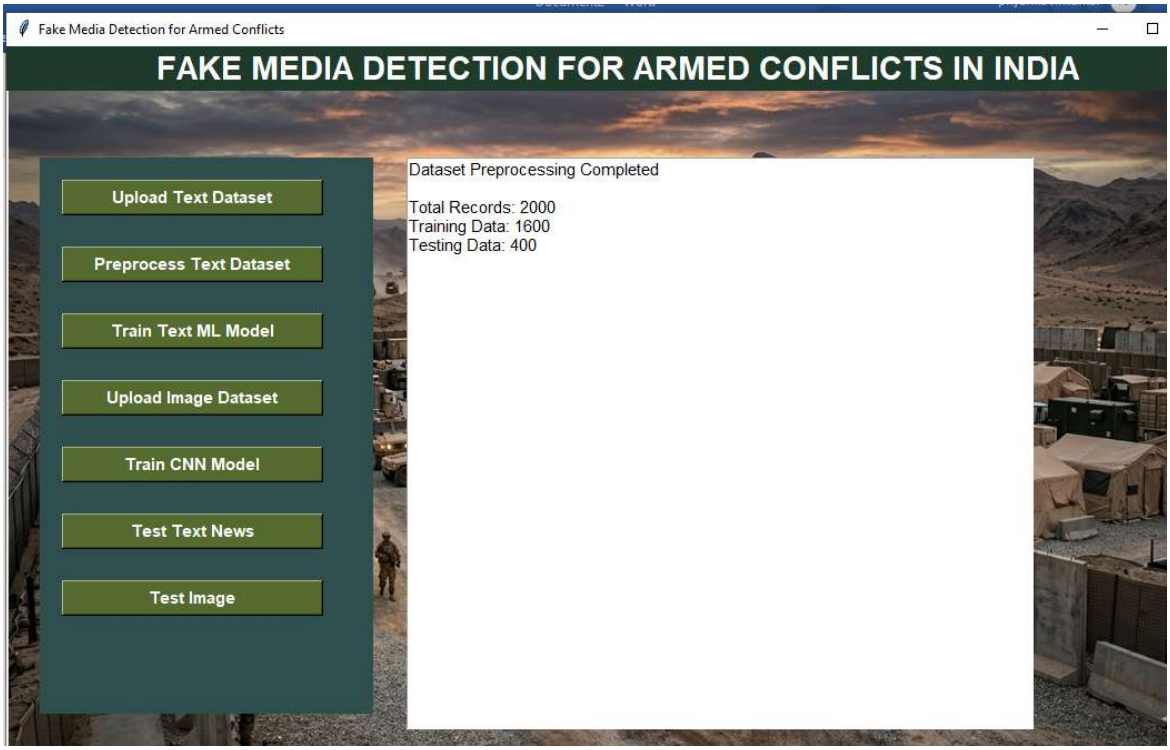
```
    panel=tk.Label(win,image=img_tk)
    panel.image=img_tk
    panel.pack()
    img_arr=image.load_img(file,target_size=(128,128))
    img_arr=image.img_to_array(img_arr)/255
    img_arr=np.expand_dims(img_arr,axis=0)
    pred=cnn_model.predict(img_arr)
    if pred[0][0]>0.5:
        text="REAL IMAGE"
        color="green"
    else:
        text="FAKE IMAGE"
        color="red"
    label=tk.Label(
        win,
        text=text,
        font=("Arial",18,"bold"),
        fg=color
    )\
    label.pack(pady=10)
# ----------------------------------
buttons=[
```

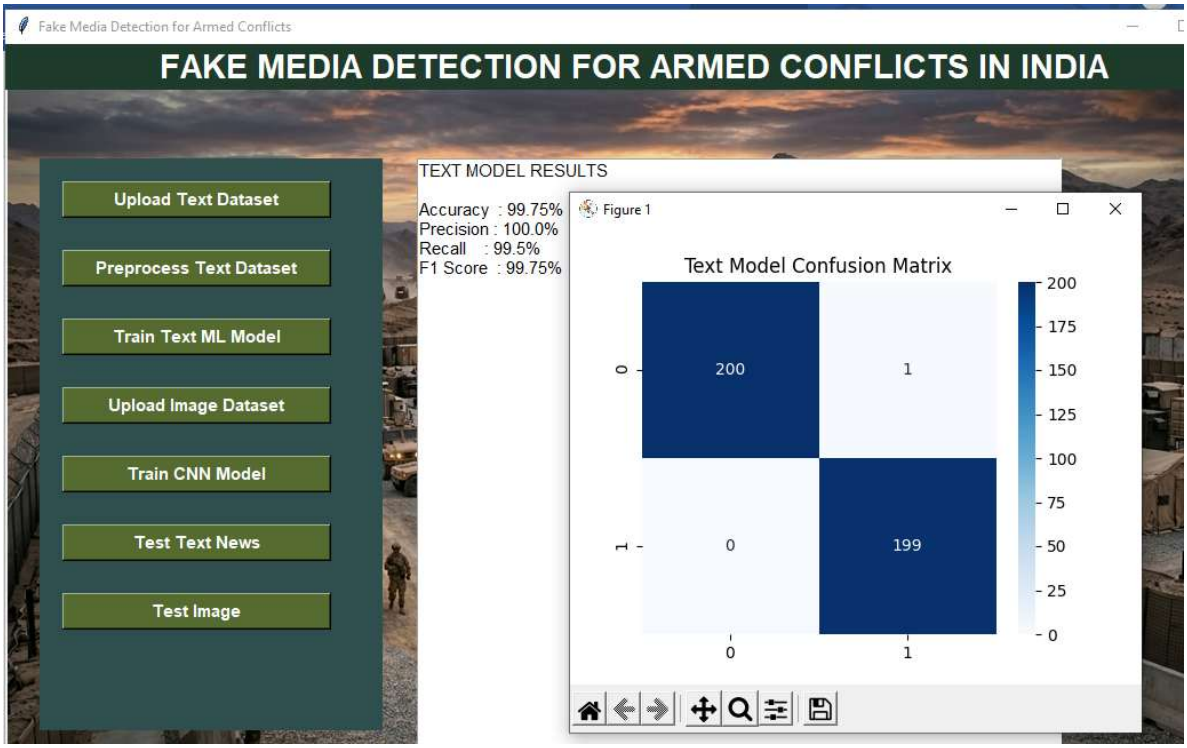
```
("Upload Text Dataset",upload_text),
("Preprocess Text Dataset",preprocess),
("Train Text ML Model",train_text_model),
("Upload Image Dataset",upload_images),
("Train CNN Model",train_cnn),
("Test Text News",test_text),
("Test Image",test_image)
]
y=20
for text,cmd in buttons:
    b=tk.Button(frame,
    text=text,
    width=25,
    bg="#556b2f",
    fg="white",
    font=("Arial",11,"bold"),
    command=cmd)
    b.place(x=20,y=y)
    y+=60
root.mainloop()
```

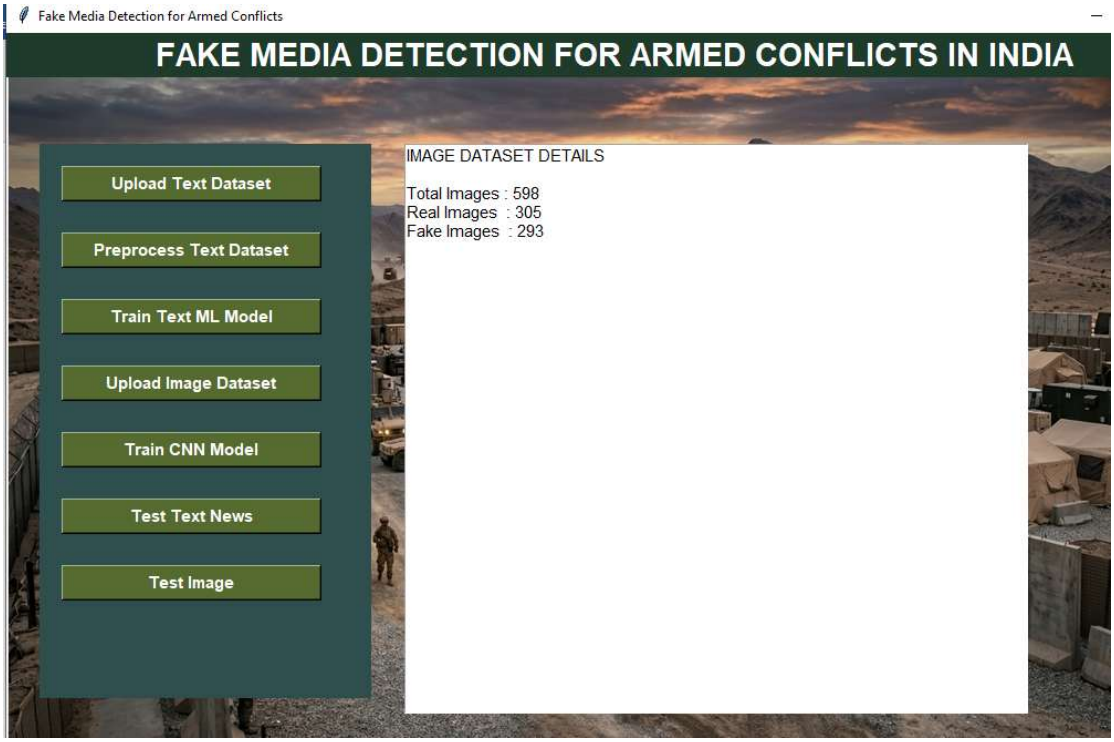
PROJECT SCREENSHOT

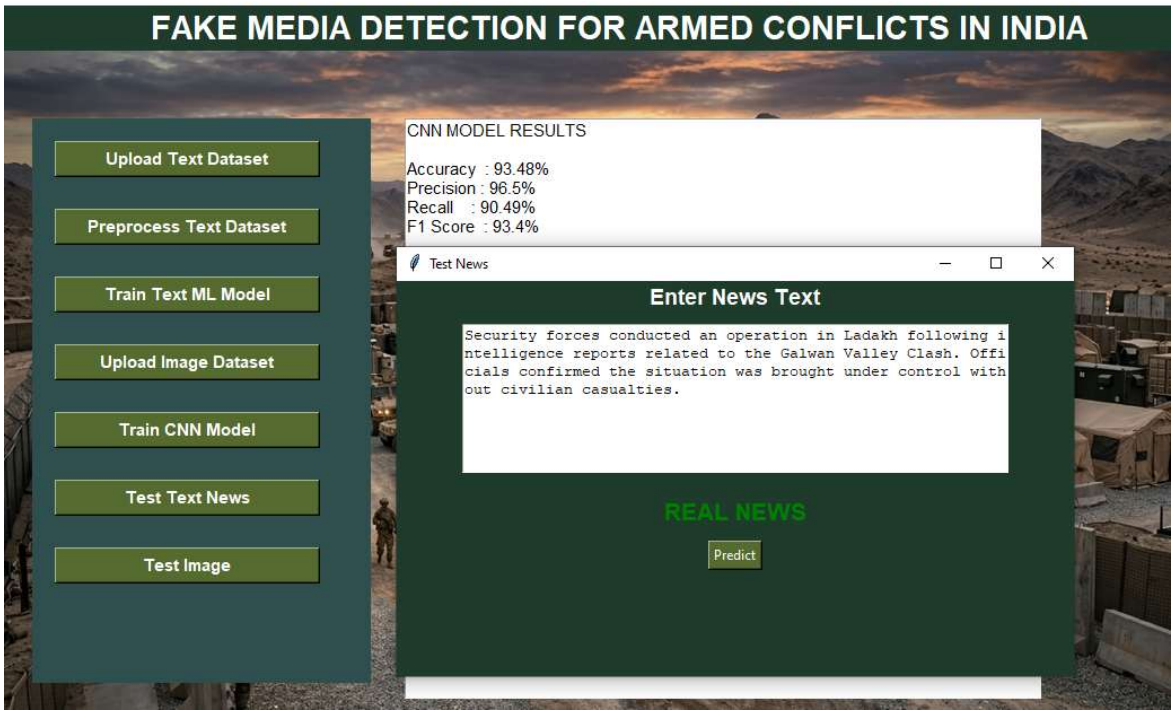


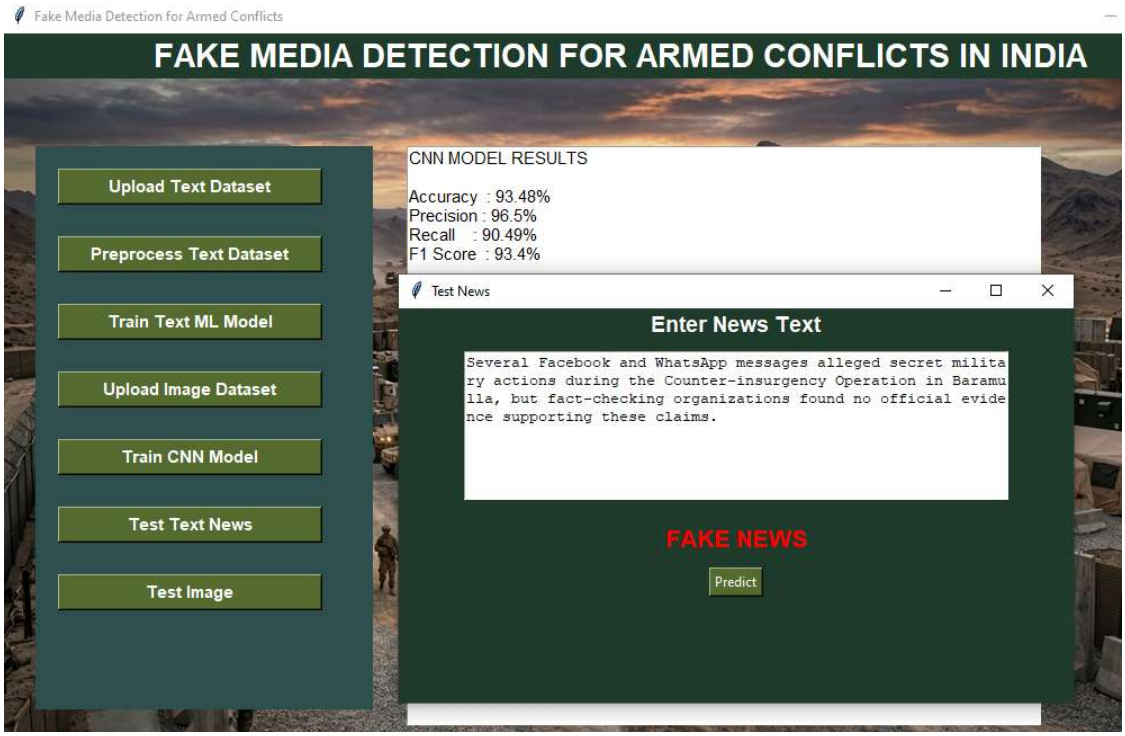


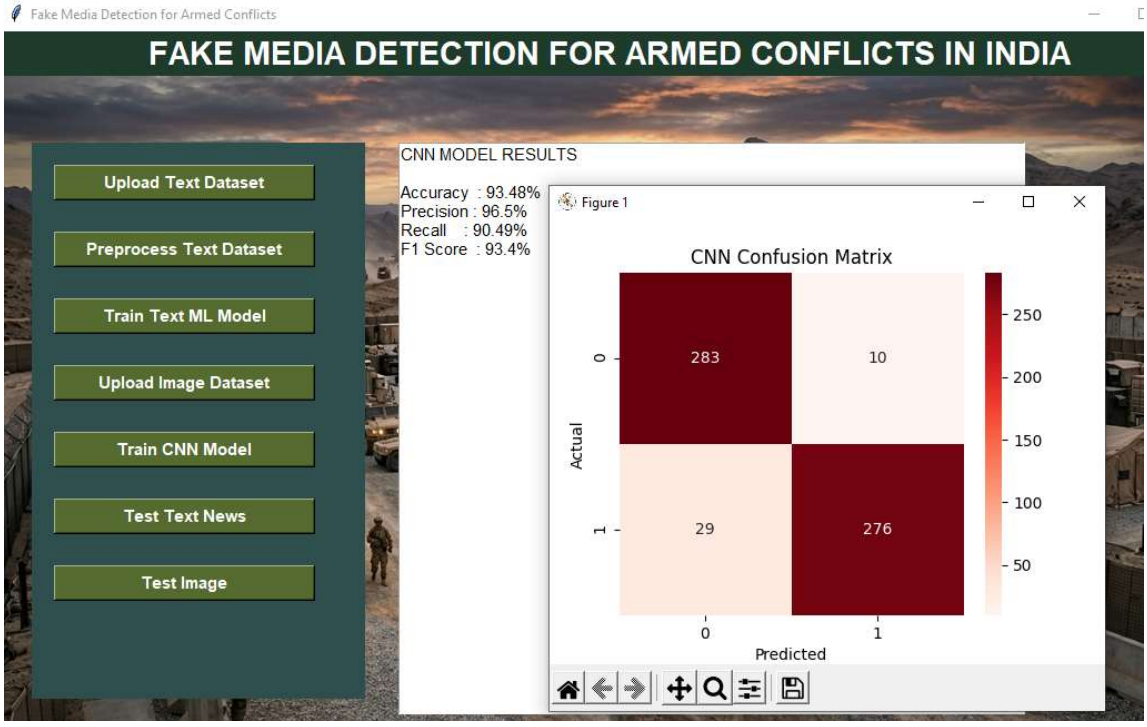

Fake Media Detection for Armed Conflicts
FAKE MEDIA DETECTION FOR ARMED CONFLICTS IN INDIA
Upload Text Dataset
Preprocess Text Dataset
Train Text ML Model
Upload Image Dataset
Train CNN Model
Test Text News
Test Image
Dataset Preprocessing Completed
Total Records: 2000
Training Data: 1600
Testing Data: 400

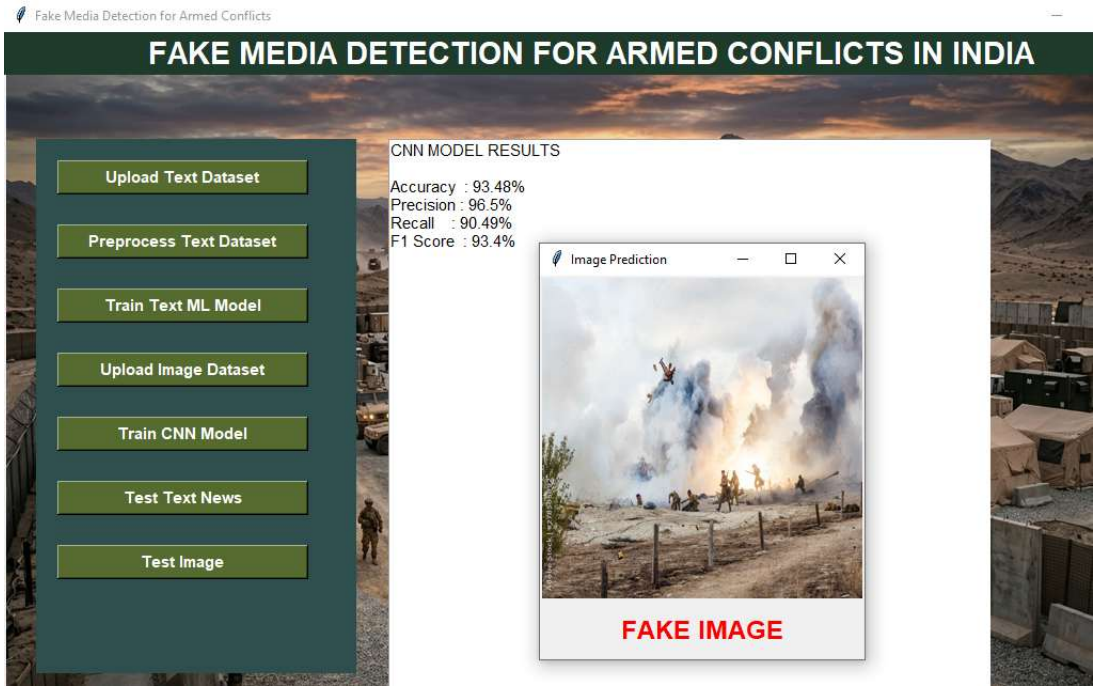

Fake Media Detection for Armed Conflicts
FAKE MEDIA DETECTION FOR ARMED CONFLICTS IN INDIA
Upload Text Dataset
Preprocess Text Dataset
Train Text ML Model
Upload Image Dataset
Train CNN Model
Test Text News
Test Image
TEXT MODEL RESULTS
Accuracy : 99.75%
Precision : 100.0%
Recall : 99.5%
F1 Score : 99.75%
Figure 1
Text Model Confusion Matrix
200
1
0
199
0
1
200
175
150
125
100
75
50
25
0


Fake Media Detection for Armed Conflicts
FAKE MEDIA DETECTION FOR ARMED CONFLICTS IN INDIA
Upload Text Dataset
Preprocess Text Dataset
Train Text ML Model
Upload Image Dataset
Train CNN Model
Test Text News
Test Image
IMAGE DATASET DETAILS
Total Images : 598
Real Images : 305
Fake Images : 293


FAKE MEDIA DETECTION FOR ARMED CONFLICTS IN INDIA
Upload Text Dataset
Preprocess Text Dataset
Train Text ML Model
Upload Image Dataset
Train CNN Model
Test Text News
Test Image
CNN MODEL RESULTS
Accuracy : 93.48%
Precision : 96.5%
Recall : 90.49%
F1 Score : 93.4%
Test News
Enter News Text
Security forces conducted an operation in Ladakh following i
ntelligence reports related to the Galwan Valley Clash. Offi
cials confirmed the situation was brought under control with
out civilian casualties.
REAL NEWS
Predict


Fake Media Detection for Armed Conflicts
FAKE MEDIA DETECTION FOR ARMED CONFLICTS IN INDIA
Upload Text Dataset
Preprocess Text Dataset
Train Text ML Model
Upload Image Dataset
Train CNN Model
Test Text News
Test Image
CNN MODEL RESULTS
Accuracy : 93.48%
Precision : 96.5%
Recall : 90.49%
F1 Score : 93.4%
Test News
Enter News Text
Several Facebook and WhatsApp messages alleged secret milita ry actions during the Counter-insurgency Operation in Baramu lla, but fact-checking organizations found no official evide nce supporting these claims.
FAKE NEWS
Predict


Fake Media Detection for Armed Conflicts
FAKE MEDIA DETECTION FOR ARMED CONFLICTS IN INDIA
Upload Text Dataset
Preprocess Text Dataset
Train Text ML Model
Upload Image Dataset
Train CNN Model
Test Text News
Test Image
CNN MODEL RESULTS
Accuracy : 93.48%
Precision : 96.5%
Recall : 90.49%
F1 Score : 93.4%
Figure 1
CNN Confusion Matrix
283
10
29
276
Actual
Predicted
0
1
250
200
150
100
50


Fake Media Detection for Armed Conflicts
FAKE MEDIA DETECTION FOR ARMED CONFLICTS IN INDIA
Upload Text Dataset
Preprocess Text Dataset
Train Text ML Model
Upload Image Dataset
Train CNN Model
Test Text News
Test Image
CNN MODEL RESULTS
Accuracy : 93.48%
Precision : 96.5%
Recall : 90.49%
F1 Score : 93.4%
Image Prediction
FAKE IMAGE


Fake Media Detection for Armed Conflicts
FAKE MEDIA DETECTION FOR ARMED CONFLICTS IN INDIA
Upload Text Dataset
Preprocess Text Dataset
Train Text ML Model
Upload Image Dataset
Train CNN Model
Test Text News
Test Image
CNN MODEL RESULTS
Accuracy : 93.48%
Precision : 96.5%
Recall : 90.49%
F1 Score : 93.4%
Image Prediction
REAL IMAGE